

Learning With AI — Topic 1

Embedded C++ Sensor Programming

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IoT Smart Parking System | NKU Spring 2026

What is Embedded C++ Programming?

"Programming that runs directly on hardware — no operating system, no screen, just code talking to physical pins."

- Embedded systems power **sensors, motors, IoT devices**
- Code runs on a **microcontroller** (ESP32, Arduino)
- Every pin is either **INPUT** or **OUTPUT**
- Timing is measured in **microseconds**
- No room for errors — wrong voltage = **damaged hardware**

Before this project: I had never written C++ or touched any hardware.

After AI guidance: I built a working IoT sensor system.

The HC-SR04 Ultrasonic Sensor

How It Actually Works

Step 1: ESP32 sends 10-microsecond HIGH pulse → TRIG pin



Step 2: Sensor fires ultrasonic sound wave



Step 3: Sound bounces off object and returns



Step 4: ECHO pin stays HIGH for duration of travel time



Step 5: ESP32 reads duration using pulseIn()



Step 6: $\text{distance} = \text{duration} \times 0.034 / 2$ (centimeters)

The Distance Formula

Breaking It Down

```
long duration = pulseIn(ECHO_PIN, HIGH);  
float distance = duration * 0.034 / 2;
```

Part	Meaning
<code>duration</code>	Time in microseconds for sound round trip
<code>× 0.034</code>	Speed of sound = 0.034 cm per microsecond
<code>÷ 2</code>	Divide by 2 because sound travels BOTH ways

Example: duration = 882 microseconds

Critical Discovery — 5V vs 3.3V

This Almost Destroyed My ESP32!

- HC-SR04 **ECHO pin OUTPUTS 5V** signal
- XIAO ESP32C6 GPIO pins only accept **3.3V maximum**
- Connecting directly = **permanent hardware damage**

The Fix — Logic Level Converter

```
HC-SR04 ECHO (5V) → Level Converter HV side
                      ↓ converts voltage
Level Converter LV side → ESP32 GPIO (3.3V)
```

GPIO Pin Control

The Three Functions I Use Every Reading

```
// Step 1 – Reset TRIG pin
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);

// Step 2 – Send 10-microsecond pulse
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

// Step 3 – Measure ECHO duration
long duration = pulseIn(ECHO_PIN, HIGH);
```

Serial Monitor Debugging

How I Tested Without a Screen

```
void setup() {  
    Serial.begin(115200); // baud rate for ESP32C6  
    Serial.println("Parking Sensor Ready!");  
}  
  
void loop() {  
    float distance = getDistance();  
    Serial.println("Distance: " + String(distance) + " cm");  
    delay(500);  
}
```

- Baud rate 9600 → Arduino Nano

The Bootloader Problem

A Hidden Arduino Setting

Upload kept failing with:

```
Error: programmer is not responding  
attempt 1 of 10: not in sync: resp=0x00
```

Root Cause:

Arduino Nano clones use **ATmega328P Old Bootloader**
but Arduino IDE defaults to the newer one.

Fix:

Complete Sensor Reading Function

```
#define TRIG_PIN 23
#define ECHO_PIN 2

void setup() {
  Serial.begin(115200);
  pinMode(TRIG_PIN, OUTPUT); // TRIG sends signal OUT
  pinMode(ECHO_PIN, INPUT);  // ECHO receives signal IN
}

float getDistance() {
  // Reset trigger
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  // Send pulse
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  // Measure echo
  long duration = pulseIn(ECHO_PIN, HIGH);

  // Calculate distance
  return duration * 0.034 / 2;
}
```

What I Learned — Summary

Concept	What I Understood
<code>pinMode()</code>	Every pin must be declared INPUT or OUTPUT
<code>pulseIn()</code>	Measures microsecond-level timing precisely
<code>delayMicroseconds()</code>	Hardware needs exact timing, not just delays
5V vs 3.3V	OUTPUT pins send voltage — must match receiver
Baud rate	Both sender and receiver must agree on speed
Old Bootloader	Hardware version affects how code uploads
Serial Monitor	Primary debugging tool for embedded systems

Key Takeaway

"Embedded programming is not just writing code — it's understanding how electricity flows through physical components. One wrong connection can permanently destroy hardware. AI helped me understand the WHY behind every connection, not just the HOW."